

BOOK MARKET INTELLIGENCE

1. BUSINESS CONTEXT

1.1 Publishing Industry Background

The global publishing industry produces millions of books across diverse genres, languages, and formats every year. As digital platforms such as Google Books, Amazon Kindle, and online bookstores continue to expand, publishers now have access to large volumes of reader engagement data, including ratings, reviews, publication trends, pricing information, and category preferences.

This growing availability of data has created opportunities for publishers, authors, and booksellers to better understand market dynamics and make informed decisions regarding content strategy, audience targeting, and inventory planning. However, the sheer volume and diversity of published books make it challenging to identify the factors that contribute to reader engagement and market success.

1.2 Business Challenge

The publishing industry generates vast amounts of data on books, authors, ratings, and reader engagement. However, identifying the factors that contribute to a book's popularity remains a challenge. Publishers need data-driven insights to understand reader preferences, evaluate category performance, track market trends, and make informed strategic decisions. This project analyzes book metadata and engagement metrics to uncover patterns that can support decision-making within the publishing ecosystem.

1.3 Key Questions Addressed

This project aims to address several business-relevant questions within the publishing industry:

- Which book categories attract the highest levels of reader engagement?
- How is book popularity distributed across the market?
- Does book length influence popularity and reader interest?
- Which languages dominate the publishing landscape?
- How have publication trends evolved over time?
- Which authors generate the highest overall engagement?

Additional analysis explores category-level popularity differences and the relationship between language and book category through statistical testing.

2. PROJECT OBJECTIVES

The primary objective of this project is to analyze book metadata and reader engagement data to uncover meaningful patterns, trends, and factors associated with book popularity within the publishing industry.

The project aims to transform raw and unstructured book data into actionable insights that can support data-driven decision-making for publishers, authors, booksellers, and market analysts.

The specific objectives of this analysis are:

- Assess the overall quality of the dataset and improve its reliability through data cleaning, preprocessing, and standardization techniques.
- Develop meaningful derived features, including a popularity score, author count, publication year, and book length categories, to enhance analytical capabilities and provide additional business context.
- Examine reader engagement patterns by analyzing ratings, popularity metrics, and category-level performance across the dataset.
- Investigate the distribution of books across languages, categories, publication years, and author structures to better understand market composition.
- Identify the categories, languages, and authors that generate the highest levels of reader engagement and market visibility.
- Explore publication trends over time to understand how the publishing landscape has evolved across different periods.
- Evaluate whether structural book characteristics, such as page count and number of authors, influence overall popularity.
- Apply statistical hypothesis testing techniques, including T-Test, ANOVA, and Chi-Square Analysis, to validate business assumptions and determine whether observed patterns are statistically significant.
- Generate data-driven market intelligence that can assist publishing stakeholders in understanding reader preferences, category performance, and emerging opportunities within the book market.
- Establish a foundation for future advanced analytics applications, including popularity prediction models, recommendation systems, and publishing trend forecasting.

By achieving these objectives, the project seeks to bridge the gap between raw publishing data and strategic business insights, enabling a deeper understanding of the factors that drive reader engagement and market success.

3. DATA UNDERSTANDING

3.1 Data Source

The dataset used in this project was obtained from a publicly available book metadata collection derived from Google Books. It contains detailed information about published books, including bibliographic details, publication information, pricing, language, ratings, and reader engagement metrics.

The dataset serves as a valuable source for analyzing publishing trends, understanding reader preferences, and exploring factors that influence book popularity across different categories and markets. The data was provided in CSV format and imported into a Python environment for preprocessing, analysis, and visualization.

3.2 Dataset Overview

The dataset contains information on thousands of books spanning multiple categories, languages, publishers, and publication years. Each record represents an individual book and includes attributes describing its content, publication details, pricing information, and reader engagement indicators.

To understand the structure and quality of the data, an initial assessment was conducted using exploratory techniques such as dataset shape inspection, column analysis, data type validation, and sample record examination.

The initial exploration provided insights into:

- The total number of records and features available for analysis.
- The distribution of numerical and categorical variables.
- Missing values and data quality issues requiring preprocessing.
- Redundant or non-analytical features that could be removed.
- Opportunities for creating additional business-relevant features.

Dataset Summary

Attribute	Value
Total Records	15,147
Total Features	21
File Format	CSV
Domain	Publishing Industry
Unit of Analysis	Individual Books

book_id	title	subtitle	authors	publisher	published_date	description	page_count	categories	average_rating	language	preview_link	
0 LR_VDQAAQBAJ	Bestsellers	The Path (bestsellers, free bestsellers, bests...	Ivan King, bestsellers	bestsellers	2017-01-04	Hear What the Critics are Saying "Wow, what an...	70.0	Young Adult Fiction	NaN	en	http://books.google.com/books?id=LR_VDQAAQBAJ&... https://play.googl	
1 WgJTDQAAQBAJ	Bestsellers	Hell: A Place Without Hope (bestseller books, ...	Ivan King, bestsellers	bestsellers	2017-01-03	Hear What the Critics are Saying Wow, very ins...	32.0	Comics & Graphic Novels	NaN	en	http://books.google.com/books?id=WgJTDQAAQBAJ&... https://play.googl	
2 4DXUDAAQBAJ	The Bestseller Code	Anatomy of the Blockbuster Novel	Jodie Archer, Matthew L. Jockers	Macmillan	2016-09-20	"What if there was an algorithm that could pre...	253.0	Business & Economics	NaN	en	http://books.google.com/books?id=4DXUDAAQBAJ&... http/	
3 yIVuDwAAQBAJ	Bestseller	A Century of America's Favorite Books	Robert McParland	Bloomsbury Publishing PLC	2018-12-15	Whether curled up on a sofa with a good myster...	335.0	Literary Criticism	NaN	en	http://books.google.com/books?id=yIVuDwAAQBAJ&... https://play.googl	
4 2JHxwAEACAAJ	Bestsellers: Popular Fiction since 1900		NaN	C. Bloom	Palgrave Macmillan	2002-07-09	This guide and reference work of all of the be...	306.0	Literary Criticism	NaN	en	http://books.google.com/books?id=2JHxwAEACAAJ&... http/

language	preview_link	info_link	isbn_13	isbn_10	list_price	currency	buyable	search_category	thumbnail
en	http://books.google.com/books?id=LR_VDQAAQBAJ&...	https://play.google.com/store/books/details?id=...	NaN	NaN	12.99	USD	True	fiction bestsellers	http://books.google.com/books/content?id=LR_VD...
en	http://books.google.com/books?id=WgJTDQAAQBAJ&...	https://play.google.com/store/books/details?id=...	NaN	NaN	9.99	USD	True	fiction bestsellers	http://books.google.com/books/content?id=WgJTD...
en	http://books.google.com/books?id=4DXUDAAQBAJ&...	http://books.google.com/books?id=4DXUDAAQBAJ&...	9.781250e+12	1250088275	NaN	NaN	False	fiction bestsellers	http://books.google.com/books/content?id=4DXUD...
en	http://books.google.com/books?id=yIVuDwAAQBAJ&...	https://play.google.com/store/books/details?id=...	9.781538e+12	1538110008	40.50	USD	True	fiction bestsellers	http://books.google.com/books/content?id=yIVuD...
en	http://books.google.com/books?id=2JHxwAEACAAJ&...	http://books.google.com/books?id=2JHxwAEACAAJ&...	9.780334e+12	0333687426	NaN	NaN	False	fiction bestsellers	http://books.google.com/books/content?id=2JHxw...

Figure 1. Sample records from the dataset.

3.3 Business-Relevant Variables

Several variables within the dataset were identified as critical for understanding publishing performance, reader engagement, and market dynamics.

Variable	Business Significance
Title	Identifies individual books within the market.
Authors	Enables analysis of author performance and collaboration patterns.
Categories	Supports category-level market analysis and popularity comparisons.
Language	Helps evaluate language-specific market trends and reader preferences.
Page Count	Used to investigate whether book length influences popularity.
Average Rating	Represents reader satisfaction and perceived book quality.
Ratings Count	Measures the volume of reader engagement.
List Price	Provides insight into pricing characteristics within the market.
Publisher	Enables examination of publishing activity across organizations.
Publication Date	Supports trend analysis across different publication periods.
ISBN-10 / ISBN-13	Unique identifies used for data validation and record reliability.

Among these variables, *Average Rating* and *Ratings Count* were combined to create a composite *Popularity Score*, which served as the primary metric for evaluating reader engagement throughout the analysis.

These business-relevant variables formed the foundation for feature engineering, market intelligence analysis, and statistical validation.

4. DATA PREPARATION AND QUALITY ENHANCEMENT

The raw dataset required several preprocessing steps to improve data quality, consistency, and analytical reliability. Initial assessment revealed missing values, duplicate records, inconsistent formatting, invalid entries, and redundant attributes that could affect the accuracy of subsequent analysis.

The data preparation process focused on enhancing the dataset's usability while preserving business-relevant information necessary for market intelligence analysis.

4.1 Data Quality Assessment

An initial review of the dataset identified several quality issues that required attention before analysis could begin. These included incomplete records, missing values across multiple fields, inconsistent data types, redundant attributes, and pricing outliers.

Key observations included:

- Missing values in fields such as subtitle, publisher, description, ISBN identifiers, and pricing information.
- Inconsistent data types across numerical and date-related variables.
- Presence of duplicate records.
- Redundant columns containing URLs and system-generated metadata with limited analytical value.
- Extreme pricing values that could distort statistical analysis.

4.2 Missing Value Treatment

Missing values were handled using a combination of imputation and standardization techniques. Numerical variables were assigned appropriate replacement values, while categorical variables were filled using meaningful default labels to preserve analytical consistency.

This approach ensured that valuable records were retained while minimizing the impact of incomplete information on the analysis.

AutoSave On Books_metadata.csv Search Aarthi Kumar

File Home Insert Draw Page Layout Formulas Data Review View Help

Clipboard Font Alignment Number Styles Cells Editing

Calibri 11 B I U Bold Italic Underline Text Color Background Color Merge & Center

General

Conditional Formatting Styles Cells Editing

T1

search_category

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	book_id	title	subtitle	authors	publisher	published_date	description	page	categories	average_rating	language	preview	link	isbn	isbn_10	list_price	currency	buyable	sea	
1	LR_VDQAA	Bestsellers The Path		Ivan King	bestsellers	04/01/2017	Hear What	70	Young Adult Fiction	0	en	http://boc	https://play.google.com/store/books/d			12.99 USD	TRUE	fict		
2	WjTDOAA	Bestsellers Hell: A		Plav Ivan King	bestsellers	03/01/2017	Hear What	32	Comics & Graphic No	0	en	http://boc	https://play.google.com/store/books/d			9.99 USD	TRUE	fict		
4	4XDUQAA	The Bestsats Anatomy		Jodie Arch Macmillan	20/09/2016	"What if I	253	Business & Economic	0	en	http://boc	https://play.google.com/store/books/d			125.0088275	FALSE	fict			
5	yIVuDAwAA	Bestseller A Century		Robert Mc Bloomsbu	15/12/2018	Whether c	335	Literary Criticism	0	en	http://boc	https://play.google.com/store/books/d			1538110008	40.5 USD	TRUE	fict		
6	2JHxWAEf	Bestsellers: Popular F.C.		Bloom Palgrave N	09/07/2002	This guide	306	Literary Criticism	0	en	http://boc	https://play.google.com/store/books/d			333687426		FALSE	fict		
7	m4yBEAA	Hype		Bestsellers Jon Helg	Nordic Ac	01/01/2015	In the wor	254	Literary Criticism	0	en	http://boc	https://play.google.com/store/books/d			31.99 USD	TRUE	fict		
8	ZW8AAAA	Bestsellers, 1979-19		Library of Congress	1	1984	58	Best Sellers	0	en	http://boc	https://play.google.com/store/books/d			ZW8AAAAAAJ&dq=fic		FALSE	fict		
9	HCoPDUZA	Mary Higg Three New		Mary Higg Wings	21/10/1996	Three New	640	Fiction	0	en	http://boc	https://play.google.com/store/books/d			517183684		FALSE	fict		
10	lK4UCAA	Bestsellers Popular F		John Sutherland	2011-08	First publi	268	Literary Criticism	0	en	http://boc	https://play.google.com/store/books/d			415611296		FALSE	fict		
11	#NAME?	The Publishers Weekly			1925		770	American literature	0	en	http://boc	https://play.google.com/store/books/d			7id=lrfsC8embo		FALSE	fict		
12	JshfCLXK	Best Sellers			1984		480	American literature	0	en	http://boc	https://play.google.com/store/books/d			JshfCLXK&dq=fiction		FALSE	fict		
13	zbT0EAAA	Richard Fifty Years		Kenneth K Bloomsbu	13/01/1988	Any future	1000	Social Science	0	en	http://boc	https://play.google.com/store/books/d			133604451	74.7 USD	TRUE	fict		
14	3fNIAQAA	Associated Advertising			1914		886	Advertising	0	en	http://boc	https://play.google.com/store/books/d			7id=3fNIAQAA		FALSE	fict		
15	H7AKAAA	Readers' Guide to Pe		Ani Lorraine Guthri	1915	An author	1478	Periodical	4	1	en	http://boc	https://play.google.com/store/books/d			7id=H7AKAAA		FALSE	fict	
16	CIQeAQAA	The Bookman			1910		690	Book collecting	0	en	http://boc	https://play.google.com/store/books/d			7id=CIQeAQAA		FALSE	fict		
17	B_OpHy8	Michigan Library Bulletin			1928		312	Libraries	0	en	http://boc	https://play.google.com/store/books/d			7id=B_OpHy8		FALSE	fict		
18	NH0IAQA	Michigan Library Bull		Michigan State Librai	1927		996		0	en	http://boc	https://play.google.com/store/books/d			7id=NH0IAQA		FALSE	fict		
19	amJm3ID	The Christian Century			1928		856	Theology	0	en	http://boc	https://play.google.com/store/books/d			7id=amJm3ID		FALSE	fict		
20	2mJHyB1	Premium Practice			1962		880	Advertising specialtie	0	en	http://boc	https://play.google.com/store/books/d			7id=2mJHyB1		FALSE	fict		
21	0IB_UGF	Library Notes		North Carolina Colleg	1926		858	Classified catalogs	0	en	http://boc	https://play.google.com/store/books/d			7id=0IB_UGF_1p		FALSE	fict		
22	1n1ngBA	Publsh. Profit. Indep		Dr. Robert Lulu	05/02/2015	Find inde	168	Business & Economic	0	en	http://boc	https://play.google.com/store/books/d			1312351969		FALSE	fict		
23	WEJIAAAA	The Recep A Study of		Meg H. Bri T&id=	1994		196	Literary Criticism	0	en	http://boc	https://play.google.com/store/books/d			7id=WEJIAAAA&dq=ficti		FALSE	fict		
24	YNLMQAA	Motivation Purpose o		Ivan King, motivatio	14/01/2017	Book Desc	98	Self-Help	0	en	http://boc	https://play.google.com/store/books/d			12.99 USD	TRUE	fict			
25	0v3PDQAA	8 Great Habits of Ext		Ivan King Valley Gro	31/01/2016	Hear What	76	Self-Help	0	en	http://boc	https://play.google.com/store/books/d			12.99 USD	TRUE	fict			

Books_metadata

Count: 15148

Figure 2. Missing value distribution across dataset features before data preparation.

4.3 Duplicate Removal

Duplicate records were identified and removed to ensure that each book was represented only once within the dataset. This step improved data integrity and prevented duplicate entries from influencing statistical results and visualizations.

4.4 Outlier Management

Outlier treatment focused primarily on book pricing data. Records containing invalid prices and extreme values were filtered to reduce distortion in pricing-related analysis.

Removing these anomalies improved the reliability of descriptive statistics and enhanced the interpretability of market trends.

4.5 Data Standardization

Several standardization procedures were applied to improve data consistency across the dataset:

- Text fields were cleaned to remove unnecessary whitespace.
- Numerical variables were converted to appropriate data types.
- Publication dates were standardized for temporal analysis.
- ISBN values were normalized for consistency.
- Language codes were mapped to descriptive language names.
- Author names were standardized where obvious naming variations existed to improve the accuracy of author-level engagement analysis.

These enhancements established a consistent analytical foundation and improved the overall quality of the dataset for subsequent feature engineering and market intelligence analysis.

5. FEATURE ENGINEERING

Feature engineering was performed to transform raw book metadata into more meaningful analytical variables that could better support market intelligence and reader engagement analysis. The newly created features enhanced the dataset's ability to uncover patterns related to popularity, authorship, publication trends, and book characteristics.

5.1 Popularity Score Framework

A composite **Popularity Score** was developed by combining a book's average rating with its ratings count. While average ratings measure reader satisfaction, ratings count reflects the volume of reader engagement. Individually, these metrics provide only a partial view of performance.

The popularity score was designed to capture both quality and engagement, creating a more comprehensive measure of a book's overall market impact. This metric served as the primary indicator of reader engagement throughout the analysis.

5.2 Author Count Analysis

Books in the dataset included both single-author and multi-author publications. To support collaboration analysis, a new variable representing the number of authors associated with each book was created.

This feature enabled the examination of whether author collaboration patterns influence popularity and reader engagement.

5.3 Book Length Segmentation

To facilitate comparative analysis, books were grouped into three categories based on page count:

- Short (<100 pages)
- Medium (100–300 pages)
- Long (>300 pages)

This segmentation enabled the evaluation of reading-length preferences and supported statistical testing of whether book length influences popularity.

5.4 Publication Year Extraction

Publication dates were transformed into a year variable to analyze publishing trends and changes in market activity over time. Collectively, the engineered features converted raw metadata into meaningful business indicators, strengthening the overall analytical framework.

	title	page_count	page_group	authors	num_authors
3	Bestseller	335.0	Long (>300 pages)	Robert McParland	1
5	Hype	254.0	Medium (100-300 pages)	Jon Helgason, Sara Kärholm, Ann Steiner	3
11	A Richard Wright Bibliography	1000.0	Long (>300 pages)	Kenneth Kinnamon, Joseph Benson, Michel Fabre,....	4
71	The Myth of Superwoman	246.0	Medium (100-300 pages)	Resa L. Dudovitz	1
82	Sisters by Choice	461.0	Long (>300 pages)	Susan Mallery	1
107	Charged	272.0	Medium (100-300 pages)	G. P. Ching	1
112	Driving Mr. Wrong Home	60.0	Short (<100 pages)	Shirley Jump	1
113	Familiar Faces	326.0	Long (> 300 pages)	Mary Roberts Rinehart	1
115	Cocoa Beans	426.0	Long (>300 pages)	Michel J. Hughes	1
117	Fictional Feminism	272.0	Medium (100-300 pages)	Kim A. Loudermilk	1

6. MARKET INTELLIGENCE ANALYSIS

Market intelligence analysis was conducted to uncover patterns in reader engagement, publishing activity, language distribution, category performance, and author influence within the book market. By examining both individual variables and relationships between key features, the analysis provides a comprehensive view of factors that shape book popularity and market dynamics.

The findings generated in this section help identify high-performing categories, dominant language markets, publication trends, and reader engagement patterns. These insights provide valuable information for publishers, authors, and booksellers seeking to make data-driven decisions regarding content strategy, audience targeting, and market positioning.

6.1 Reader Engagement Landscape

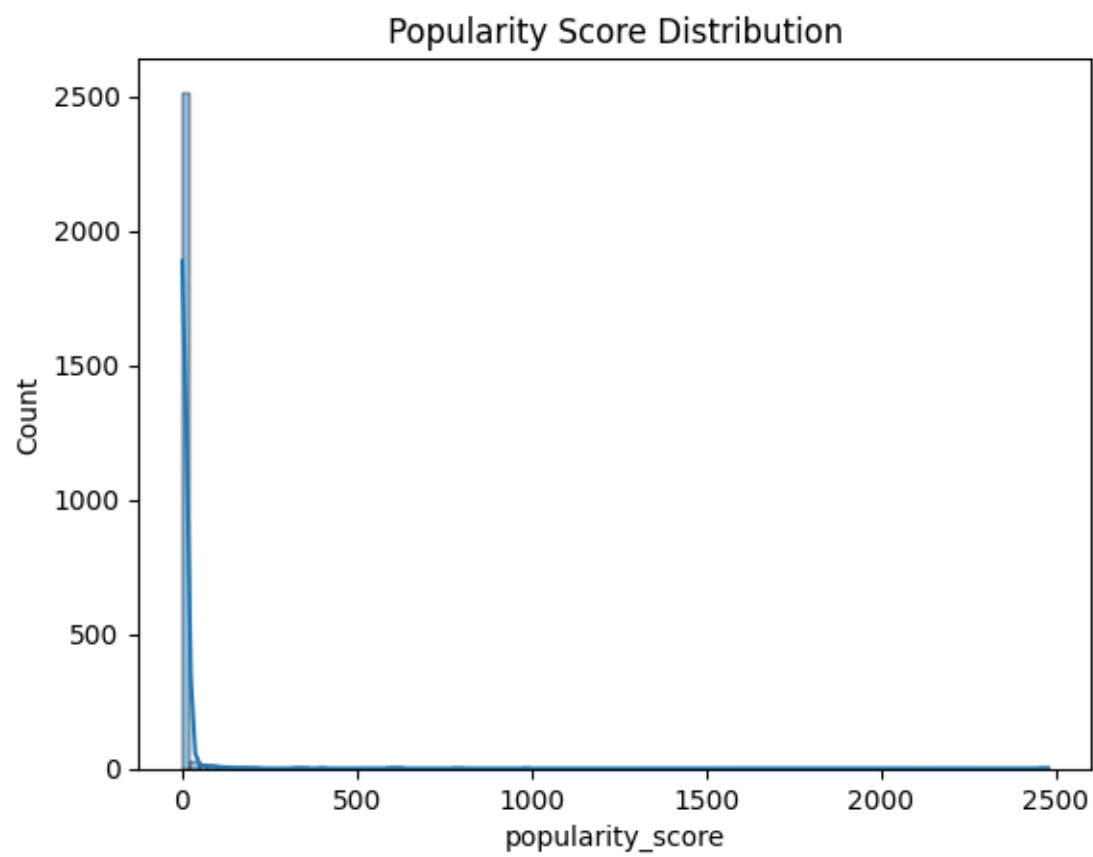


Figure 3. Distribution of popularity scores across books.

Reader engagement is highly concentrated among a small number of books. Most books exhibit relatively low popularity scores, while a few titles achieve exceptionally high levels of engagement, creating a strongly right-skewed distribution.

6.2 Language Market Analysis

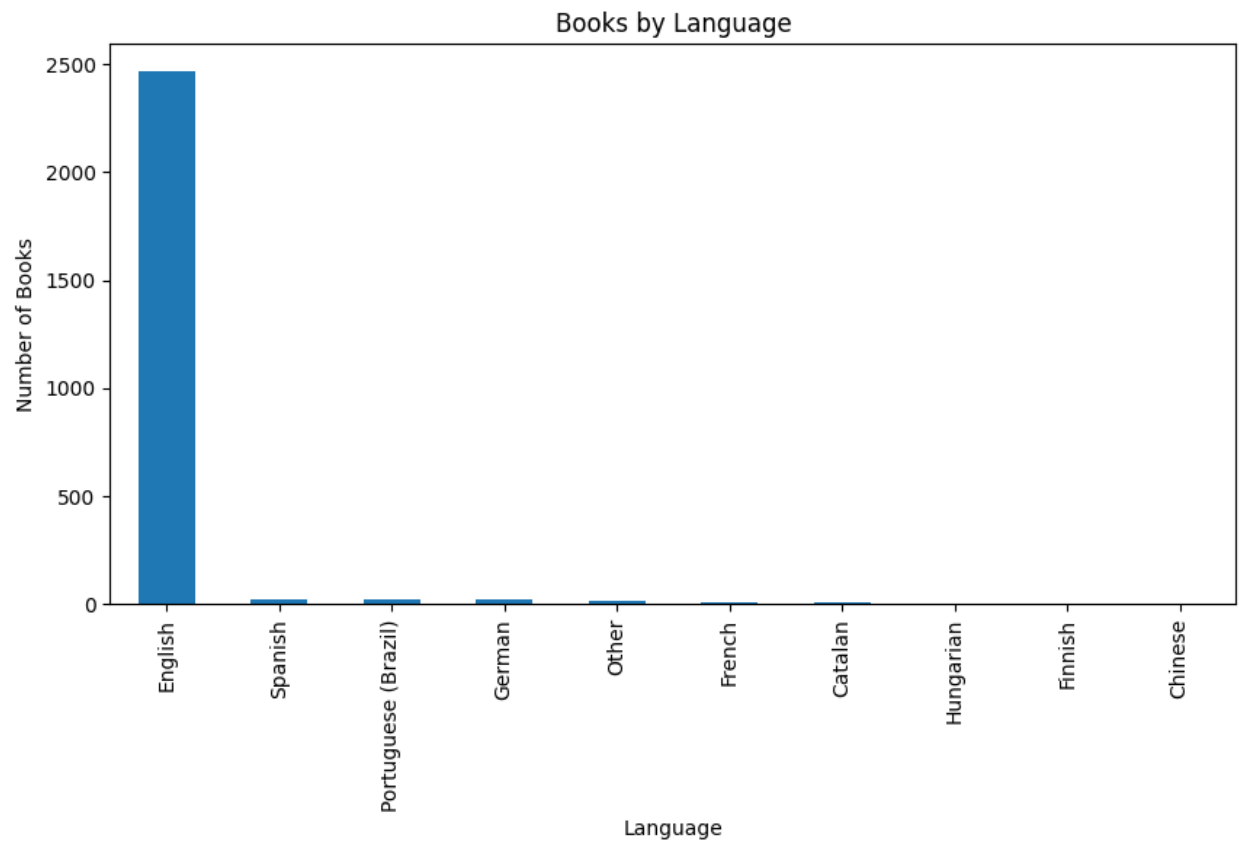


Figure 4. Distribution of books across language markets.

English dominates the publishing landscape with 2,470 books, representing the overwhelming majority of records. Other languages such as Spanish, Portuguese (Brazil), and German account for only a small fraction of the dataset, indicating that reader engagement and publishing activity are largely concentrated within English-language markets.

6.3 Category Performance Analysis

Fiction emerged as the dominant category, achieving a popularity score of 8,824.5, significantly outperforming all other genres. Juvenile Fiction ranked second with 1,283.0, followed by History and Business & Economics. This suggests that reader engagement is heavily concentrated in fiction-oriented content.

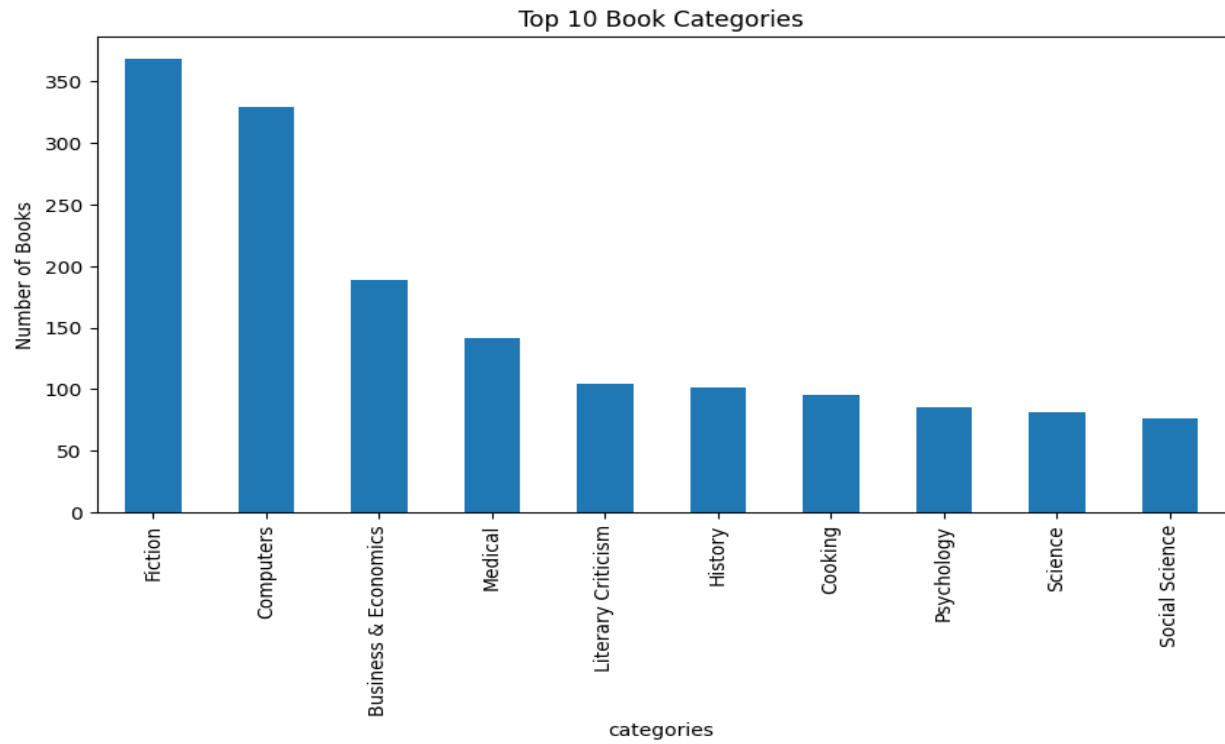


Figure 5. Most represented book categories within the dataset.

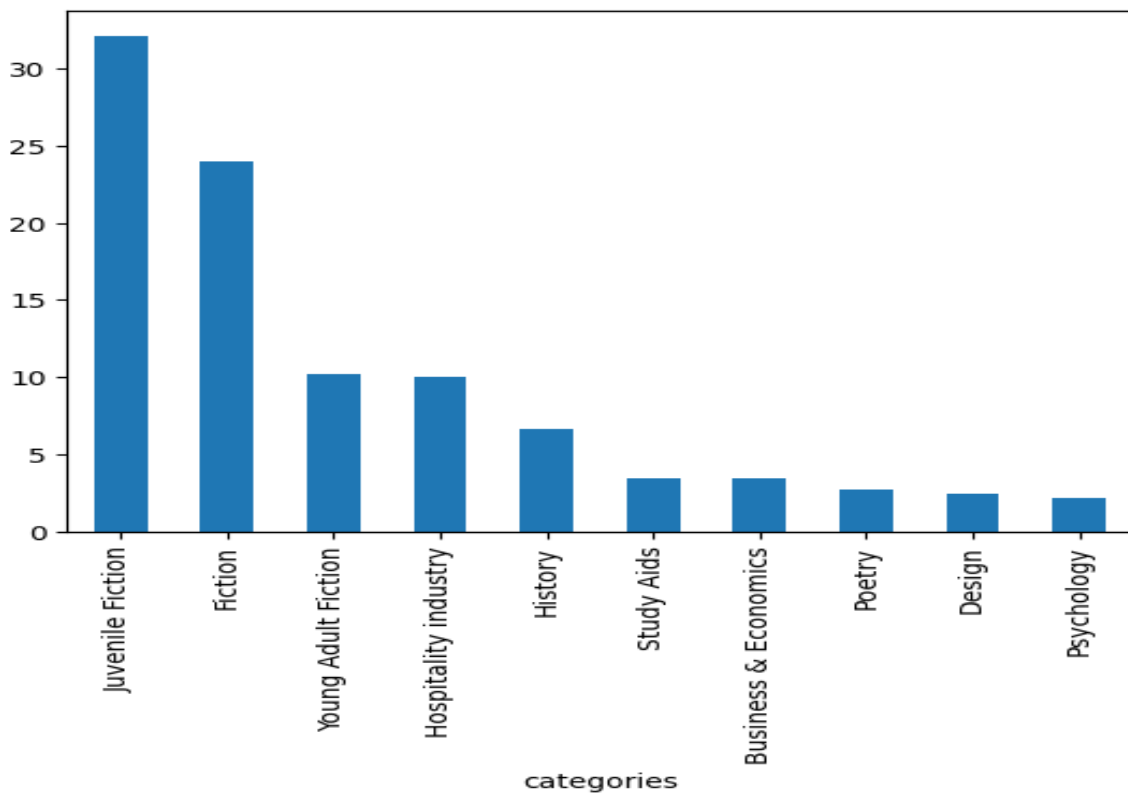


Figure 6. Categories generating the highest reader engagement.

6.4 Publication Trend Analysis

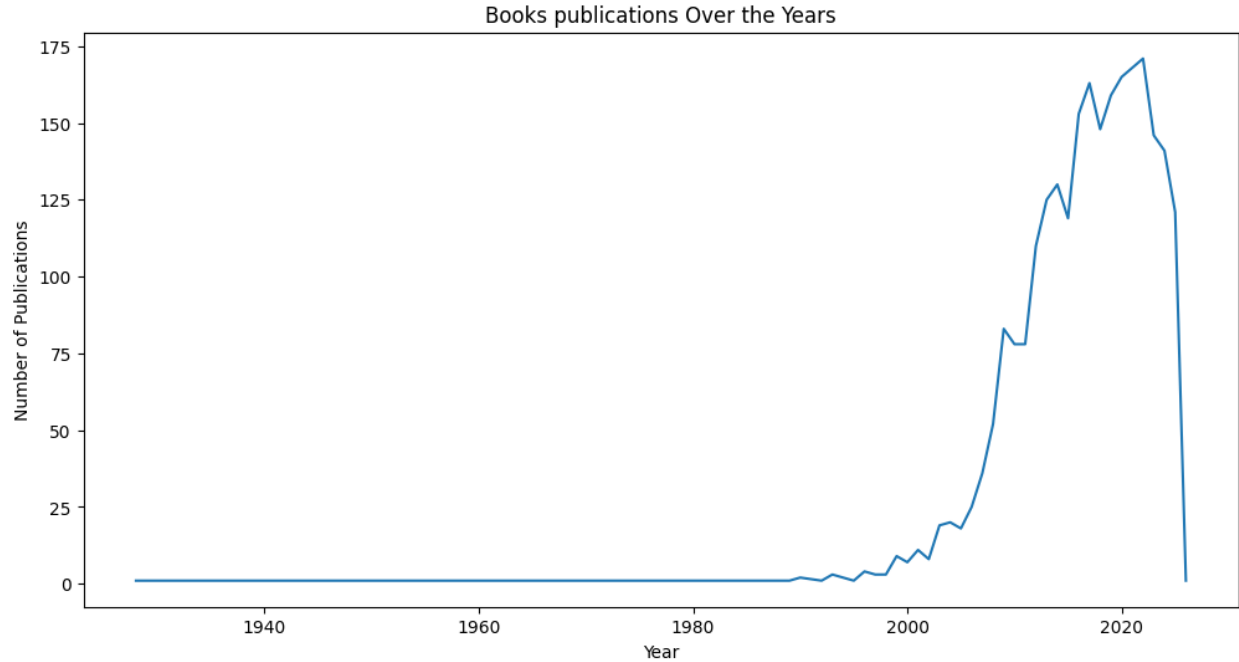


Figure 7. Publication volume trends across years.

Publication trends were analyzed to understand how publishing activity has evolved over time. Examining publication volume and reader engagement across years helps identify growth patterns, shifts in market activity, and long-term industry trends.

6.5 Author Performance Analysis

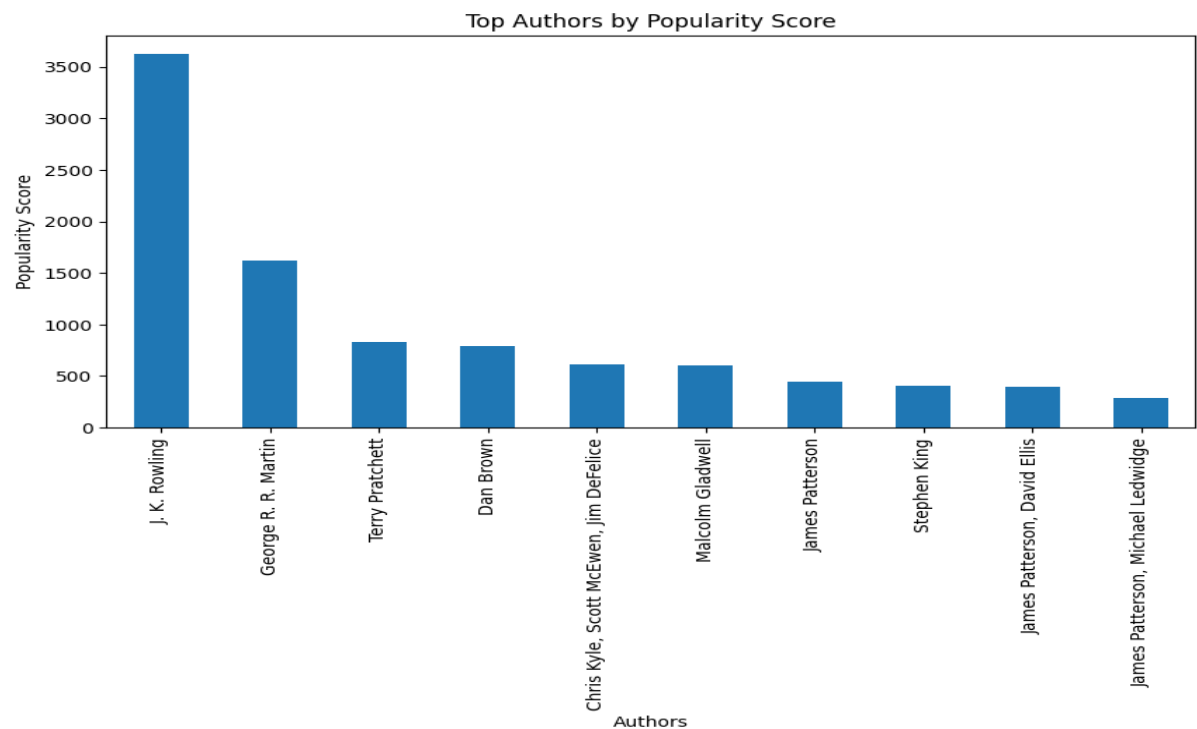


Figure 8. Authors generating the highest reader engagement.

J. K. Rowling achieved the highest popularity score (3,621.5), followed by George R. R. Martin (1,620.0) and Terry Pratchett (826.5). The results highlight the significant influence of established authors in driving reader engagement and market visibility.

6.5.1 Correlation Analysis Insights

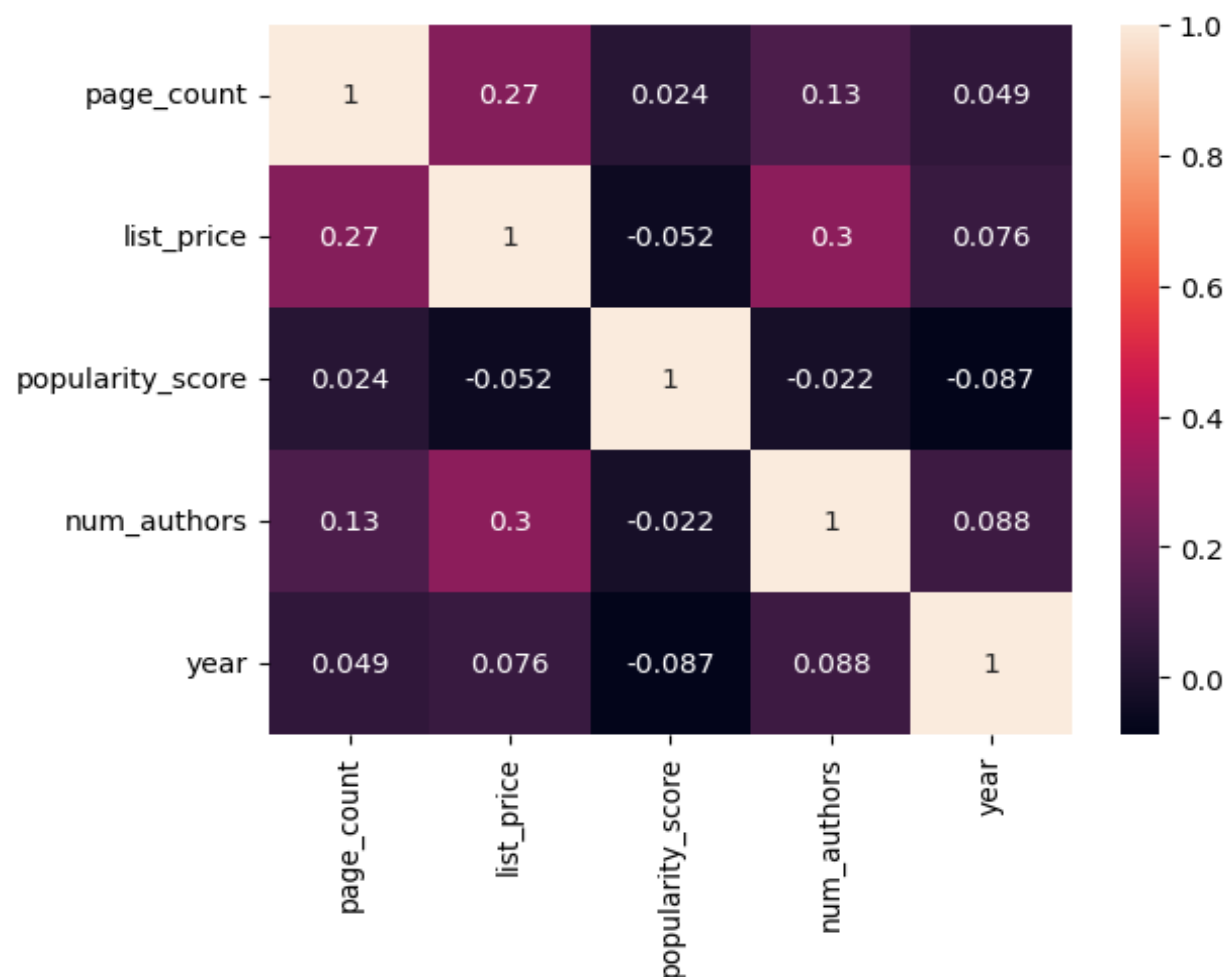


Figure 9. Relationship between popularity score, page count, and author count.

Correlation analysis revealed weak relationships among all numerical variables. Popularity score showed negligible correlations with page count (0.024), publication year (-0.087), and number of authors (-0.022), indicating that these factors have limited influence on reader engagement.

The strongest observed correlation was between list price and number of authors (0.30), followed by page count and list price (0.27). However, both relationships remain relatively weak and do not indicate strong dependence between the variables. Overall, reader engagement appears to be influenced more by content-related factors such as category appeal and author reputation than by structural book characteristics.

7. STATISTICAL VALIDATION OF BUSINESS QUESTIONS

While market intelligence analysis provides valuable insights into publishing trends and reader engagement patterns, statistical testing is necessary to determine whether observed relationships are statistically significant. To validate key business assumptions, hypothesis testing techniques were applied to assess the influence of book characteristics on popularity and examine relationships between important market variables.

The statistical validation framework focused on three key business questions:

- Does book length significantly influence reader engagement?
- Do popularity levels differ across book categories?
- Is there a relationship between language and book category?

To answer these questions, an Independent Sample T-Test, One-Way ANOVA, and Chi-Square Test of Independence were performed. These tests provide quantitative evidence that supports or challenges the patterns identified during the market intelligence analysis.

7.1 Does Book Length Influence Popularity?

Independent Sample T-Test

An Independent Sample T-Test was conducted to determine whether book length significantly influences reader engagement. Books were grouped into short and long categories based on page count, and their popularity scores were compared.

Business Question

Does book length significantly influence book popularity?

Hypotheses

- Null Hypothesis (H_0): There is no significant difference in popularity between short and long books.
- Alternative Hypothesis (H_1): There is a significant difference in popularity between short and long books.

Metrics	Value
T-Statistic	-1.060
P-Value	0.290

Interpretation

Since the p-value (0.290) is greater than the significance level of 0.05, the null hypothesis cannot be rejected. The analysis indicates that there is no statistically significant difference in popularity between short and long books.

Business Insight

Despite common assumptions that longer books generate higher reader engagement, the analysis found no statistically significant relationship between book length and popularity. Highly popular books were observed across both short and long page-count segments, suggesting that factors such as content quality, author reputation, category relevance, and audience interest are more important drivers of engagement than book length alone.

7.2 Do Categories Perform Differently?

One-Way ANOVA

A One-Way Analysis of Variance (ANOVA) was performed to determine whether reader engagement differs significantly across book categories.

Business Question

Do certain book categories generate significantly higher popularity than others?

Hypotheses

- Null Hypothesis (H_0): Mean popularity is equal across all categories.
- Alternative Hypothesis (H_1): At least one category has a significantly different mean popularity.

Metrics	Value
F-Statistic	4.291
P-Value	0.00189

Interpretation

Since the p-value (0.00189) is lower than the significance level of 0.05, the null hypothesis is rejected. This indicates that popularity differs significantly across categories.

Business Insight

The statistical findings support the category performance analysis conducted earlier in the study. Fiction emerged as the highest-performing category with a popularity score of **8,824.5**, significantly outperforming all other genres. Juvenile Fiction ranked second with **1,283.0**, followed by History (**675.0**) and Business & Economics (**655.0**). These results demonstrate that category selection plays a critical role in reader engagement and market performance. Publishers can leverage these insights to prioritize high-demand genres and optimize content investment strategies.

7.3 Are Language and Category Related?

Chi-Square Test of Independence

A Chi-Square Test of Independence was conducted to evaluate whether language and book category are associated.

Business Question

Is there a significant relationship between language and book category?

Hypotheses

- Null Hypothesis (H₀): Language and category are independent.
- Alternative Hypothesis (H₁): Language and category are associated.

Metrics	Value
Chi-Square Statistic	104.48
P-Value	1.36×10^{-8}

Interpretation

Since the p-value is significantly lower than 0.05, the null hypothesis is rejected. The analysis provides strong evidence that language and category are not independent.

Business Insight

The findings indicate that language markets exhibit distinct category preferences. This observation aligns with the language distribution analysis, where English-language books

dominated the dataset with **2,470 titles**, while Spanish (**23**), Portuguese (Brazil) (**22**), and German (**20**) represented considerably smaller segments. The results suggest that publishing activity and reader preferences vary across language markets, highlighting opportunities for market segmentation, localization strategies, and targeted publishing initiatives.

7.4 Statistical Validation Summary

The statistical analysis produced three important findings:

- Book length was not found to have a statistically significant influence on popularity ($p = 0.290$).
- Reader engagement differs significantly across book categories ($p = 0.00189$).
- Language and category exhibit a strong statistical association ($p = 1.36 \times 10^{-8}$).

Overall, the results indicate that category selection and language-specific market preferences have a greater influence on reader engagement than structural characteristics such as page count. These findings reinforce the importance of category strategy, audience targeting, and market segmentation in driving publishing success and maximizing reader engagement.

8. KEY FINDINGS

Finding 1: Fiction Dominates Reader Engagement

Fiction emerged as the highest-performing category with a popularity score of **8,824.5**, significantly outperforming all other genres. This indicates that reader engagement is heavily concentrated in fiction-oriented content, highlighting strong and sustained demand within this segment.

Finding 2: English-Language Books Dominate the Market

English-language books accounted for **2,470 titles**, far exceeding all other language groups in the dataset. This suggests that publishing activity and reader engagement are primarily concentrated within English-speaking markets.

Finding 3: Author Reputation Plays a Significant Role in Engagement

Author-level analysis revealed that **J. K. Rowling** achieved the highest popularity score (**3,621.5**), followed by **George R. R. Martin** (**1,620.0**) and **Terry Pratchett** (**826.5**). The findings indicate that established authors have a substantial influence on reader engagement and market visibility.

Finding 4: Category Selection Significantly Influences Popularity

Statistical testing confirmed that popularity differs significantly across book categories (**ANOVA p = 0.00189**). This finding demonstrates that genre choice is an important factor influencing reader engagement and overall market performance.

Finding 5: Book Length Does Not Significantly Affect Popularity

The Independent Sample T-Test found no statistically significant difference in popularity between short and long books (**p = 0.290**). This suggests that book length alone is not a reliable predictor of reader engagement.

Finding 6: Language and Category Are Closely Related

The Chi-Square Test revealed a significant association between language and category (**p = 1.36×10^{-8}**). This indicates that category preferences vary across language markets and that publishing strategies should consider linguistic audience differences.

Finding 7: Structural Book Characteristics Have Limited Influence on Engagement

Correlation analysis showed very weak relationships between popularity score and page count (**0.024**), publication year (**-0.087**), and number of authors (**-0.022**). These results suggest that reader engagement is driven more by content relevance, category appeal, and author reputation than by structural book attributes.

9. STRATEGIC RECOMMENDATIONS

Based on the findings derived from market intelligence analysis and statistical validation, the following strategic recommendations are proposed for publishers, authors, and publishing organizations seeking to enhance reader engagement and market performance.

Recommendation 1: Prioritize High-Performing Categories

The analysis identified Fiction and Juvenile Fiction as the strongest-performing categories in terms of reader engagement. Publishers should prioritize investment in these genres through content acquisition, marketing initiatives, and audience development strategies to capitalize on existing demand.

Recommendation 2: Adopt Category-Specific Publishing Strategies

Statistical testing confirmed significant differences in popularity across categories. Rather than adopting a uniform publishing approach, organizations should tailor content strategies according to category performance, reader preferences, and market demand within each genre.

Recommendation 3: Focus on Content Quality Over Book Length

Since book length was not found to significantly influence popularity, publishers should focus on content quality, storytelling effectiveness, and audience relevance rather than structural characteristics such as page count.

Recommendation 4: Leverage Strong Author Brands

Author performance analysis demonstrated that established authors generate substantially higher reader engagement. Publishers should leverage successful literary brands through targeted promotions, strategic partnerships, and long-term audience engagement initiatives.

Recommendation 5: Implement Language-Specific Market Strategies

The significant relationship between language and category suggests that reader preferences vary across language markets. Publishers should adopt localized content strategies, language-specific marketing campaigns, and audience segmentation approaches to improve market reach and engagement.

10. CONCLUSION

This project transformed raw Google Books metadata into actionable market intelligence through data cleaning, feature engineering, exploratory analysis, and statistical validation. The analysis revealed that reader engagement is strongly influenced by category selection and language-specific market preferences, while structural characteristics such as book length have limited impact on popularity.

Fiction emerged as the dominant category, English-language books accounted for the majority of publishing activity, and established authors generated the highest levels of reader engagement. Statistical testing further confirmed significant differences in popularity across categories and a strong relationship between language and category.

Overall, the project demonstrates how data analytics and statistical techniques can be applied to uncover meaningful publishing trends, support evidence-based decision-making, and generate valuable insights for publishers, authors, and booksellers.

11. FUTURE ENHANCEMENTS

- Develop machine learning models to predict book popularity.
- Build personalized book recommendation systems.
- Incorporate reader reviews and sales data for deeper market insights.
- Perform sentiment analysis on book descriptions and reviews.
- Expand analysis to publisher-level performance evaluation.
- Explore regional and language-specific publishing trends.
- Develop interactive dashboards for real-time market monitoring.